

A Matter of Routine

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As a primary-schooler, I enjoyed arithmetic, and found it fun to compute sums, factorise large numbers, and so on. I know that in academic mathematics, we like to distance ourselves from these activities a bit, since it's good to clarify that arithmetic skills aren't a prerequisite for doing interesting mathematics or good research, but I'd be lying if I said it wasn't a gateway of sorts for me. Still, aside from a few moments here and there, by the time that I hit secondary school, it was reasonably clear that this was not going to be a key component of my mathematical character, and that there were other more important things to move on to. I had gained various things from it (including a vague sense of personal familiarity with prime factorisations up to a given size, which persists to this day), but it would not feature in my day with any regularity.

In late secondary school, I had my first encounters with calculus, and acquainted myself thoroughly with integrals, sums, and series. I developed a zeal for solving integrals, evaluating infinite series, and various things in between. The bank of problems to work through is richer than for finite sums and factorisations, and the set of tools to throw at them correspondingly larger. On top of that, sometimes these problems fail to admit 'closed-form solutions', which, while upsetting at the time, turned out to be rather eye-opening in the longer term. In any case, I honed these skills for some time, built up some familiarity, and ultimately came to a similar conclusion to the one I had for arithmetic: I'm very glad that I can make these calculations, and they're likely to be tangibly useful for tasks in front of me for a few more years yet, but they will only take me so far.

As I moved into university and research, plenty of other vignettes like this unfolded, in which I encountered new areas filled with examples and learned to speak the language of those examples formally. You find your way around these topics, acting out the moves repeatedly until you feel more and more at home. Often, you'll find a codification of these strategies in a way that distills the essence of the topic and reduces it to something which can be reliably resolved by a predictable set of steps. You might go from verifying convexity of functions by hand to 'disciplined convex programming', from verifying reversibility of Markov kernels by hand to identifying 'extension-involution pairs', or from proving convergence of a convex optimisation algorithm by hand to 'PEP-certification' of the algorithm by methods of semidefinite programming.

Countless other instances doubtless exist. My experience is that the first phase of immersing oneself in a new topic and its canon is rarely felt as a 'pointless exercise in some obsolete task', even in hindsight, and instead persists as a fond memory of when I 'really learned' the subject, or at least felt myself to be a part of it. That aspects of the topic begin to feel 'routine' after the fact is a natural culmination, and is arguably what we strive for in much of mathematics. We clear the path forwards for others, so that they might follow it more smoothly than we did, and so that we might focus our efforts on those parts which remain rather rougher around the edges.

At the time of writing, the maelstrom of developments surrounding the field of mathematics interacts with this process in a fascinating way. From where I sit, there is an undeniable sense that many aspects of our subject which have long been dense, time-consuming, or otherwise unwieldy might quickly become 'routine'. Some of the usual growing pains around this passage from 'unwieldy' to 'routine' remain, though they have themselves arguably also been attenuated by the same developments; the dynamics of complaining at a teacher until they provide you with a nicer proof are rather more favourable in the present day!

A novel challenge perhaps then concerns whether to engage with these growing pains at all. In the past, borrowing results from an unfamiliar field would require either an investment of one's own time, the borrowing of a friendly expert's time, the boldness to cite an unfamiliar reference without the usual level of understanding, or some mixture of all three. Each of these avenues is arguably easier to take than it has ever been, and I can see that the choice of which road to go down is rather personal (and even topic-dependent). Faced with a certain number of terms in a Taylor expansion, or a certain number of arrows in a diagram, the temptation to 'turn to the calculator' looms large.

Still, there are good reasons to insist on making the journey oneself. The familiarity that one feels after making sense of a topic is more than just a warm, fuzzy feeling in one's chest. In absorbing mathematical concepts at our own pace, we develop comfort and creativity. It becomes easier to identify connections passively and explore them actively; we 'speak the language' without having to pick up the dictionary for each word. Moreover, this approach offers some hope of a more certain path forwards. The hierarchical structure of knowledge and expertise gives us good cause to develop and refine our foundations as we go, avoiding a 'collapsing house of cards' wherein we stack more and more complex and unfamiliar topics on top of one another in pursuit of an isolated answer. Respecting our mathematical tools in this way offers some insurance for their future development by ourselves and others.

That arithmetic skills would eventually become 'routine' would perhaps have been clear to me even as a young child. That more advanced mathematics would also become 'routine' should also have been clear to me as a university student (though the scale and pace of this would of course have been rather more difficult to anticipate). In all cases, the journey through these a posteriori 'routine' landscapes was a part of my development as a mathematician, and their passage into the canon is a triumph of the subject.

If I am lucky, by the time that I finish my career as a mathematician, many more of my favourite topics and results will be seen as 'routine' by the next generation of mathematicians, who will have journeyed through a much cleaner landscape to understanding. They will undoubtedly be better off for it, and I will scarcely be worse off for having tried to help things along.